The Running Least Common Multiple of a Three-Prime Smooth Prefix

Lean-checked structure for Erdős Problem #269

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Abstract

Let p, q, r be pairwise distinct primes and let $\mathrm{L}(x)$ denote the least common multiple of all $\{p, q, r\}$-smooth numbers not exceeding x. We prove in Lean that

$$\mathrm{L}(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor} \qquad (x \geq 1),$$

so the running least common multiple is the product of the three maximal pure prime powers below the cutoff, and is at most x^3. Consequently it is constant on each cell where the three integer logarithms are constant; when exactly one logarithm advances by one it is multiplied by exactly the corresponding prime; and the first n positive powers in the three channels, together with the common origin 1, give exactly $3n+1$ distinct jump values. We prove an exact finite normal form in which a rectangular lattice sum of reciprocal running values is grouped by its genuine height, with each coefficient the cardinality of the corresponding fibre, and a uniform bound $9\,\#\mathcal{S} \leq (j+3)^2$ on the cardinality of a smooth exponent shell whose sorted height coordinates sum to j. At $\{2, 3, 5\}$ we exhibit the smallest exact obstruction to separating the kernel as a product, the two sides being $1/60$ and $1/12$.

Problem #269 is open and nothing here decides it. We prove no irrationality and no transcendence statement in any three-prime case. The residue-escape route is represented only by its final step, a finite contradiction whose escape hypothesis is assumed and not proved; we state that hypothesis explicitly, and it is the obligation that survives.

1 The problem

Let P be a finite set of primes with $|P| \geq 2$, and let $a_1 < a_2 < \cdots$ enumerate the positive integers all of whose prime factors lie in P. Erdős Problem #269 asks whether

$$\sum_{n \geq 1} \frac{1}{[a_1, \ldots, a_n]}$$

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is irrational, where $[a_1, \dots, a_n]$ is the least common multiple [1, p. 65][2, p. 106]. Numbering and status follow Bloom's catalogue [4]. The problem is open for every finite P with $|P| \geq 2$.

Three things are known and fix the shape of the question. The restriction $|P| \geq 2$ is necessary: for $P = \{p\}$ the enumeration is $a_n = p^{n-1}$, so $[a_1, \dots, a_n] = p^{n-1}$ and the sum is $p/(p-1)$, which is rational. For infinite P the sum is always irrational, which Erdős calls a simple exercise [2, p. 106]. And in a letter of 1 January 1973 he recorded that he could prove irrationality once duplicate summands are removed [3].

That last remark is the one this note is closest to. The running least common multiple is not injective in n: it is constant along stretches of the enumeration and changes only at certain points. Removing duplicate summands means summing over the distinct values rather than over n. Sections 3 and 4 make that reindexing exact, in the case $|P| = 3$: they identify where the value is constant, by exactly what factor it changes when it changes, and what multiplicity each distinct value carries. They do not recover the irrationality proof Erdős reported for the de-duplicated sum, and they say nothing about the original sum.

Throughout, p, q, r are pairwise distinct primes. Call n *smooth* when $n = p^i q^j r^k$ for some $i, j, k \geq 0$; this is the smooth lattice value. For $x \geq 1$ write

$$\mathrm{L}(x) = \operatorname{lcm}\{\, n \leq x : \ n \text{ smooth} \,\}, \qquad \mathrm{H}(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor},$$

the running least common multiple and the pure-power height, where $\lfloor \log_b x \rfloor$ is the integer logarithm, the largest e with $b^e \leq x$. Since $a_1, \dots, a_n$ are exactly the smooth numbers up to a_n, we have $[a_1, \dots, a_n] = \mathrm{L}(a_n)$, and the summands of the problem are the reciprocals of L along the enumeration. The reciprocal of the height at a smooth point is the lattice kernel

$$\mathrm{K}(i, j, k) = \frac{1}{\mathrm{H}(p^i q^j r^k)}.$$

Scope of this note, and relation to prior work. We treat $|P| = 3$ throughout, writing $P = \{p, q, r\}$, and the smallest instance is $\{2, 3, 5\}$. Nothing here is specific to three generators for its own sake; three is where the coordinates stop separating, which Theorem 6.1 makes exact at the smallest instance. The statement of Problem #269 has been formalised before, as a conjecture with an unfilled proof, in the *formal-conjectures* collection. That is a formal statement of the question. The declarations described below are propositions about the objects the question is posed over, and no claim of priority is made for any of them.

Status of everything below. *Status.* The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Statement	Status	Treatment here
Irrationality in any three-prime case	Open	Not proved.
$\mathrm{L}(x) = \mathrm{H}(x)$	Proved here	Theorem 2.1; needs the primes distinct.
$\mathrm{H}(x) \le x^3$	Proved here	Proposition 2.2.
Constancy on logarithmic cells	Proved here	Theorem 3.1.
Single-coordinate jump ratios	Proved here	Theorem 3.2.
Exactly $3n + 1$ jump values	Proved here	Theorem 3.3.
Height-fibre normal form	Proved here; finite	Theorem 4.1.
Infinite limit of that expansion	Not proved	Section 4.
Shell multiplicity $9\,\#\mathcal{S} \le (j+3)^2$	Proved here	Theorem 5.3.
Kernel is not separable at $\{2, 3, 5\}$	Proved here	Theorem 6.1.
Residue escape occurs	Assumed, not proved	Section 7.
Finite contradiction consuming it	Proved here	Theorem 7.1.

Reading map. Section 2 identifies the running value. Sections 3 and 4 develop its jump structure and the finite normal form, Section 5 bounds fibre multiplicity, and Section 6 records the obstruction to separating the kernel. Section 7 is the one place where an unproved hypothesis appears, and it is marked as such throughout. Linked phrases open the corresponding Lean declaration at the pinned source revision 4eb4c8ecfa0a.

Keywords. irrationality; least common multiple; smooth numbers; lattice sums; Lean 4.
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2 The running value is a product of pure powers

The smooth numbers up to x are indexed by the exponent triples (i, j, k) with $i \le \lfloor \log_p x \rfloor$, $j \le \lfloor \log_q x \rfloor$, $k \le \lfloor \log_r x \rfloor$ and $p^i q^j r^k \le x$: the coordinate bounds make the index set finite, and the last condition is the actual constraint. This is the smooth prefix index set. Both conditions matter: the coordinate box is strictly larger than the prefix, since a product of three large pure powers can exceed x while each factor does not.

Theorem 2.1 (the running least common multiple)**.** *Let p, q, r be pairwise distinct primes and $x \ge 1$. Then* $\mathrm{L}(x) = \mathrm{H}(x)$.

Proof. For divisibility in one direction, every smooth $n \le x$ has exponents bounded by the corresponding integer logarithms, so $n \mid \mathrm{H}(x)$, and hence $\mathrm{L}(x) \mid \mathrm{H}(x)$. For the other, the three pure powers $p^{\lfloor \log_p x \rfloor}$, $q^{\lfloor \log_q x \rfloor}$ and $r^{\lfloor \log_r x \rfloor}$ are themselves smooth numbers not exceeding x, so each divides $\mathrm{L}(x)$. Distinct primes have coprime powers, so their product divides $\mathrm{L}(x)$ as well. The two divisibilities give equality. $\square$

Formalised as the running-lcm identity, from the divisibility into the height and the three membership statements for the pure powers, namely the first, second, and third pure-power memberships.

The hypothesis that the primes are pairwise distinct is used exactly once, in the coprimality step, and it is not decorative: without it the three pure powers need not have coprime orders and their product need not divide the least common multiple. The identity is what makes every later statement about L computable from three integer logarithms.

Proposition 2.2. *For every $x \geq 1$, $\mathrm{H}(x) \leq x^3$.*

Proof. Each of the three factors is a power of its base not exceeding x. □

Formalised as the cubic majorant, which needs no primality. The exponent 3 is the number of generating primes, and the bound is the reason the reciprocal kernel is comparable to x^{-3} rather than to x^{-1}.

3 Cells, jumps, and their count

Say that x and y lie in the same *logarithmic cell* when $\lfloor \log_b x \rfloor = \lfloor \log_b y \rfloor$ for each of $b = p, q, r$: the cell relation.

Theorem 3.1 (constancy on cells)**.** *If $x, y \geq 1$ lie in the same logarithmic cell, then $\mathrm{L}(x) = \mathrm{L}(y)$. The same holds for the kernel at two smooth points whose values lie in one cell.*

Proof. Immediate from Theorem 2.1, since H depends on x only through the three integer logarithms. □

Formalised as the cell constancy of the running value, the cell constancy of the height, and the cell constancy of the kernel. The running value therefore changes only when one of the three logarithms changes, and the next theorem says by exactly how much.

Theorem 3.2 (single-coordinate jump ratios)**.** *Let $x, y \geq 1$. If $\lfloor \log_p y \rfloor = \lfloor \log_p x \rfloor + 1$ while the other two logarithms agree, then $\mathrm{L}(y) = p\,\mathrm{L}(x)$; and similarly with q or r in place of p.*

Proof. By Theorem 2.1 both sides are heights, and one factor of the height gains one in its exponent while the others are unchanged. □

Formalised as the first, second, and third coordinate steps, over the corresponding statements for the height alone, which need no primality: the first height step, the second, and the third.

The jump values are therefore the pure prime powers, and they do not collide across channels.

Theorem 3.3 (jump count)**.** *Let $n \geq 0$. The set of the first n positive powers of p, of q, and of r has exactly $3n$ elements, and adjoining the common origin 1 gives exactly $3n + 1$.*

Proof. Within one channel the powers $b, b^2, \ldots, b^n$ are distinct because $b \geq 2$. Across two channels a common value would be a positive power of two distinct primes, which unique factorisation forbids. Finally 1 is not a positive power of any prime. □

Formalised as the positive jump count and the jump count with the origin, over the channel cardinality, the channel disjointness, and the exclusion of the origin; the channels themselves are the positive power sets. The exponent 0 is omitted from each channel because it is the shared initial value, which is why the origin is counted once rather than three times.

4 The height-fibre normal form

Grouping a lattice sum by the value of the running least common multiple turns it into a sum over heights whose coefficients are multiplicities. We prove this exactly, on a finite rectangular box.

Write $B(h_p, h_q, h_r)$ for the box of exponent triples with $i \le h_p$, $j \le h_q$, $k \le h_r$, the exponent box, and for a height H write $F(H)$ for the set of points of the box whose running height equals H, the height fibre.

Theorem 4.1 (finite normal form)**.** *For every box,*

$$\sum_{(i,j,k)\in B} \mathrm{K}(i,j,k) = \sum_{H} \frac{\#F(H)}{H},$$

the outer sum ranging over the heights actually attained on B.

Proof. Partition B into the fibres of the height map. On $F(H)$ every summand is $1/H$ by definition of the kernel, so the fibre contributes $\#F(H)/H$. □

Formalised as the height-fibre normal form, over the fibre sum, with the height of a lattice point given by the point height.

Together with Theorem 3.1 this is the finite core of the ordered prime-power jump expansion: the value is constant on cells, the cells are indexed by heights, and the coefficient of a height is the number of lattice points it collects. The passage to the infinite sum, and the explicit ordering of the pure powers that would make the expansion a series in the jumps, are not proved here. Theorem 4.1 is an identity between two finite sums, and it is stated over the full box rather than the smooth prefix, so it is not a statement about L at a cutoff.

The tail dynamics that such an expansion would feed is recorded in the same module as a one-step map, $\tau(b,d,s) = b(s-d)$, the variable-base tail step, together with the rewriting that names its expanded form. No orbit of this map is analysed.

5 Shell multiplicity

A tail estimate needs to know how many lattice points can share a short multiplicative interval. Write $\mathcal{S}$ for the set of exponent triples in a box $B(h_p, h_q, h_r)$ whose smooth value lies in $[\lambda, \eta)$, the smooth exponent shell.

Lemma 5.1 (uniqueness in a short interval)**.** *Let $b \ge 1$ and suppose $\eta \le b\lambda$. If $b^a w$ and $b^{a'} w$ both lie in $[\lambda, \eta)$, then $a = a'$.*

Proof. If $a < a'$ then $\eta \le b\lambda \le b \cdot b^a w = b^{a+1} w \le b^{a'} w < \eta$, which is impossible; the case $a > a'$ is symmetric. □

Formalised as the short-interval uniqueness. The hypothesis is that the interval has multiplicative width at most b; then one coordinate is determined by the other two, and projecting it away is injective.

Proposition 5.2. *If $\eta \le r\lambda$ then $\#\mathcal{S} \le (h_p+1)(h_q+1)$; if $\eta \le p\lambda$ then $\#\mathcal{S} \le (h_q+1)(h_r+1)$.*

Formalised as the third-coordinate projection and the first-coordinate projection.

Theorem 5.3 (quadratic multiplicity bound)**.** *Suppose $\eta \le r\lambda$ and $h_p \le h_q \le h_r$ with $h_p + h_q + h_r = j$. Then*

$$9\,\#\mathcal{S} \le (j+3)^2.$$

Proof. By Proposition 5.2, $\#\mathcal{S} \le (h_p+1)(h_q+1)$. It therefore suffices to prove that $a \le b \le c$ with $a+b+c = j$ gives $9(a+1)(b+1) \le (j+3)^2$. From $a \le b \le c$ we get $a + 2b \le j$, so it is enough that $9(a+1)(b+1) \le (a+2b+3)^2$; writing $b = a+d$ with $d \ge 0$, the difference of the two sides is $d(3a+4d+3) \ge 0$. □

Formalised as the quadratic shell bound, over the sorted quadratic estimate. The constant 9 is the square of the number of generating primes and appears because the bound is the arithmetic–geometric comparison for a sum of three sorted coordinates; sorting is a hypothesis, not a normalisation, since the shell itself is not symmetric in the three bases.

The bound is uniform in λ and η subject to the width condition, and it is stated for the actual filtered shell rather than for a lattice model of it. It is an input to a tail estimate and is not itself one: no series is bounded here.

6 The kernel does not separate

A recurring hope is to write the three-prime kernel as a product of functions of the separate coordinates, which would reduce the problem to one-dimensional criteria. At the smallest instance this fails, and the failure is exact.

Theorem 6.1 (non-separability at $\{2,3,5\}$)**.** *With $(p,q,r) = (2,3,5)$,*

$$\mathrm{K}(0,0,0)\ \mathrm{K}(1,1,0) = \tfrac{1}{60} \ne \tfrac{1}{12} = \mathrm{K}(1,0,0)\ \mathrm{K}(0,1,0).$$

Proof. The four values are $\mathrm{K}(0,0,0) = 1$, $\mathrm{K}(1,0,0) = 1/2$, $\mathrm{K}(0,1,0) = 1/6$ and $\mathrm{K}(1,1,0) = 1/60$, computed from $\mathrm{H}(1) = 1$, $\mathrm{H}(2) = 2$, $\mathrm{H}(3) = 2\cdot 3 = 6$ and $\mathrm{H}(6) = 4\cdot 3\cdot 5 = 60$. □

Formalised as the non-separation witness, over the four exact values, the origin, value at two, value at three, and value at six. The height at 6 is 60 rather than 6 because the maximal pure powers below 6 are 4, 3 and 5: the running least common multiple at a smooth cutoff sees powers of the other primes that the cutoff itself does not contain. That is the mechanism behind the non-separation.

A nonzero two-by-two minor rules out writing the kernel as $f(i)g(j)h(k)$ on this box, and hence rules out any argument that first separates the coordinates and then treats them one at a time. It does not bound the rank of the kernel from below beyond one, and it is not an independence or irrationality statement.

7 The residue-escape route and what it assumes

One strategy for the three-prime case runs as follows. Suppose the value is rational with denominator C. The jump expansion produces an integer tail state that is positive and bounded by a polynomial window; if one can force the state to be congruent modulo C to a least positive residue lying strictly above that window, the two facts are incompatible and the assumed rationality fails. The final step is elementary and we record it; the step that would make the strategy an argument is the production of the escaping residue, and it is not proved.

Say that a residue *escapes* the window when bound $<$ residue $\leq C$, the escape condition.

Theorem 7.1 (the finite contradition)**.** *Let c, C, bound, residue be natural numbers with $0 < c$ and $c \leq$ bound, and suppose $c \equiv$ residue $\pmod{C}$. If the residue escapes the window, the hypotheses are contradictory. Equivalently, under the same congruence, residue $\leq$ bound.*

Proof. From $c \leq$ bound $<$ residue $\leq C$ we get $c < C$, so c and residue are each their own least residue modulo C unless residue $= C$, in which case its least residue is 0; in either case the congruence forces $c =$ residue or $c = 0$, and both contradict $c \leq$ bound $<$ residue and $0 < c$. $\square$

Formalised as the finite contradiction and the contrapositive form.

We state plainly what this is. Theorem 7.1 is two lines of arithmetic about residues and contains no information about primes, smooth numbers, or least common multiples. It is the trivial end of the strategy, recorded so that the nontrivial end is visible as the only thing missing. The hypothesis that an escaping residue occurs, for every putative denominator and beyond every level, is exactly the open producer; assuming it is not progress towards it.

8 Complements and further questions

Problem #269 is open. Three obligations survive the results above, in decreasing order of directness.

Problem 8.1 (uniform residue escape)**.** Show that under the rational carry recurrence some prime-power channel carries a nonzero residue beyond every finite level, uniformly in the putative denominator. Theorem 7.1 then closes the argument; Theorem 5.3 supplies the polynomial window the escape must clear.

Problem 8.2 (a genuinely three-dimensional analytic theorem)**.** Prove an analytic statement for the phase cocycle attached to three independent logarithmic coordinates, of the kind available in one dimension. By Theorem 6.1 no separation into one-dimensional factors is available even at $\{2, 3, 5\}$.

Problem 8.3 (unbounded denominator exclusion)**.** Replace a finite denominator exclusion by a family whose exclusion bound tends to infinity. A finite computation excludes all denominators below its bound and none above it, so no single such computation, at any size, bears on irrationality.

What the formalised material changes is the description of the object, not its status. The running least common multiple is now an explicit function of three integer logarithms; its jumps are exactly the pure prime powers; a lattice sum groups exactly by height with multiplicities bounded quadratically; and the coordinates provably do not separate. None of this produces the escaping residue, and the problem remains open.

Statements and declarations

Artefact and data availability. The pinned formal-source revision contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. This manuscript provides navigation rather than proof authority.

Declaration of generative AI use. Large-language-model agents were used throughout development to draft and revise prose, formal proofs, and software. The author set the objectives and acceptance criteria, selected and reviewed the claims, and approved the published version. The author assumes responsibility for the accuracy, interpretation, and presentation of the work. Generative systems are production tools; they are not authors and supply no independent authority. Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision 4eb4c8ecfa0a. The structural material is in the first module and the residue consumer in the second. Three distinctions are worth carrying into the source. The height statements hold for arbitrary bases, while the statements about L need the three primes to be distinct. The normal form of Section 4 is an identity between finite sums over a rectangular box, not over a smooth prefix. And the escape condition of Section 7 is a hypothesis of its theorem, not a result.

References

[1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. 28, Geneva, 1980, p. 65.

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[3] P. Erdős, letter to the editor (written 1 January 1973), Fibonacci Quart. **12** (1974), p. 335.

[4] T. F. Bloom, *Erdős Problem #269*, `erdosproblems.com/269`, accessed 22 July 2026.